Wine Time
A Multivariate Look into Wine Type and Quality

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February, 2026

1 Introduction

The dataset used in this study by Cortez et al. (2009) contains physicochemical measurements and quality ratings for wines from the Vinho Verde region of northwestern Portugal. Compiled from laboratory analyses and sensory evaluations by professional tasters, the data include 11 continuous chemical measurements, a wine-type indicator (red or white), and a sensory quality rating on a 0–10 scale.

This report addresses two questions:

1. What physicochemical characteristics differentiate red wines from white wines?
2. To what extent do these chemical variables help explain wine quality?

We approach these questions beginning with exploratory data analysis (EDA), correlation analysis, and finally principal component analysis (PCA).

2 Data Overview

After loading the dataset with 6497 observations and 13 variables, we identified 1177 duplicate rows. Because the duplicates also repeated the quality rating, they did not appear to represent the same wine evaluated multiple times; therefore, the duplicates were removed. The resulting dataset contains 5320 unique wines: 3961 white wines and 1359 red wines.

2.1 Variables

The 11 physicochemical variables measure various aspects of acidity levels, sugar content, sulfur dioxide, density, pH, sulphates, and alcohol contained in wines. Each of these variables is accounted for in the wine-making process, contributing to the final product. Listing the effects of the included variables from Cortez et al. (2009) and extra information provided from Waterhouse et al. (2016):

- **fixed_acidity**: Primarily tartaric acid, influencing flavor and inhibiting microbial growth through its low pH
- **volatile_acidity**: Acetic acid, a natural byproduct from yeast during alcoholic fermentation. Leads to an unsavory vinegar taste if in too high concentrations

- **citric_acid**: Adds a zesty, citrus note to wine
- **residual_sugar**: Remaining after fermentation
- **chlorides**: Contributes to the salt content often attained from the terroir of the region and natural absorption rates from grape varietals
- **free_sulfur_dioxide**: Prevents oxidation
- **total_sulfur_dioxide**: Includes free sulphur and that coming from yeast during fermentation. Also responsible for the "contains sulfites" label found on many wines due to their concentration over 10 mg/L
- **density**: Near that of water
- **pH**: Overall acidity of the wine
- **sulphates**: Antimicrobial additive
- **alcohol**: % volume alcohol content

2.2 Distributional Characteristics

Kurtosis values from the univariate histograms in Figure 1 reveal that most variables are right-skewed with heavy tails, particularly residual sugar, chlorides, and sulfur dioxide. Preliminary analysis of these outliers suggests that high concentrations of volatile acidity and chlorides generally reflect poorly on wine quality, with the weak negative correlations in Figure 2A. This supports the idea that balance in chemical composition may correlate with wine quality. In contrast to the more skewed predictors, pH and alcohol have approximately symmetric, near-normal distributions with skewness values close to zero.

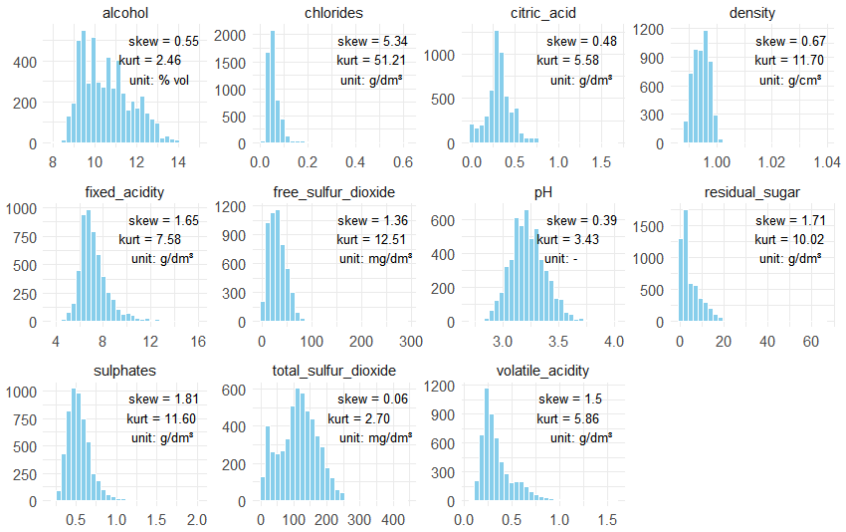


Figure 1: Distribution of the predictor variables (p = 11) for red and white wines and their unit measurements. Most chemical attributes exhibit substantial right-skew and heavy-tailed behavior, supported by listed skewness and kurtosis values, while some are comparatively symmetric.

We opted to rely on visual assessment along with kurtosis and skewness measures when determining normality of features, as the Shapiro-Wilk test is very sensitive to large sample sizes.

Finally, because the predictors vary substantially in scale, e.g., total sulfur dioxide ranges from 6 to 440 mg/dm^3, whereas density spans only 0.99 to 1.04, and because PCA assumes linearity and is sensitive to inconsistent scales, all variables were standardized by centering and scaling prior to analysis.

To address the imbalance of quality ratings and ensure stable group comparisons, we recoded quality into the following clusters:

- **Bad**: 0-4
- **Average**: 5-7
- **Great**: 8-10

This yielded 236 bad, 153 great, and 4931 average wines.

2.3 Multivariate Normality Check

We applied Mardia's multivariate skewness and kurtosis test, which strongly rejected MVN for skew (p-value ≈ 0). Because PCA is robust to MVN violations but sensitive to outliers and differing scales, this result supports our decision to center and scale all variables.

3 Correlation Structure

Because the scale of variables differs widely, we opted for correlation heatmaps rather than covariance, since predictors with larger values (e.g., total_sulfur_dioxide) would dominate. To avoid Simpson's paradox when aggregating red and white wines together, we generated heatmaps for all wines, only white wines, and only red wines.

3.1 Observations from Correlation Heatmaps

As previously discussed, the correlations for the heatmap for all wines (Figure 2A) reveals few notable correlations beyond expected physical phenomena. For example, density is positively correlated with residual sugar ($r \approx 0.52$) and strongly negatively correlated with alcohol ($r \approx -0.67$), reflecting the expected physical effect of sugar increasing the density of wine (similar to its effect in water, a similarly dense liquid) while alcohol decreases density. Acidity-related variables show similar patterns: fixed acidity and citric acid are positively correlated ($r \approx 0.33$), while pH is negatively correlated with both fixed acidity and citric acid ($r \approx -0.27$ and $r \approx -0.34$, respectively), consistent with the fact that greater acidity corresponds to lower pH.

When red and white wines are examined separately, these relationships become more distinct and align with known enological differences, directly addressing our first research question on characterizing wine types. White wines exhibit a very strong correlation ($r \approx 0.82$) between residual sugar and density, consistent with their generally sweeter profiles (Figure 2C). Meanwhile, red wines show stronger linkages among fixed acidity, citric acid, and density ($r \approx 0.67$ each), resulting in a lower mean pH of 0.11, consistent

with traditional knowledge that red wines generally have lower pH than their white counterparts. These contrasts highlight the distinct chemical structures between the types of wines and directly motivate our first research question about the characterization of red versus white wines.

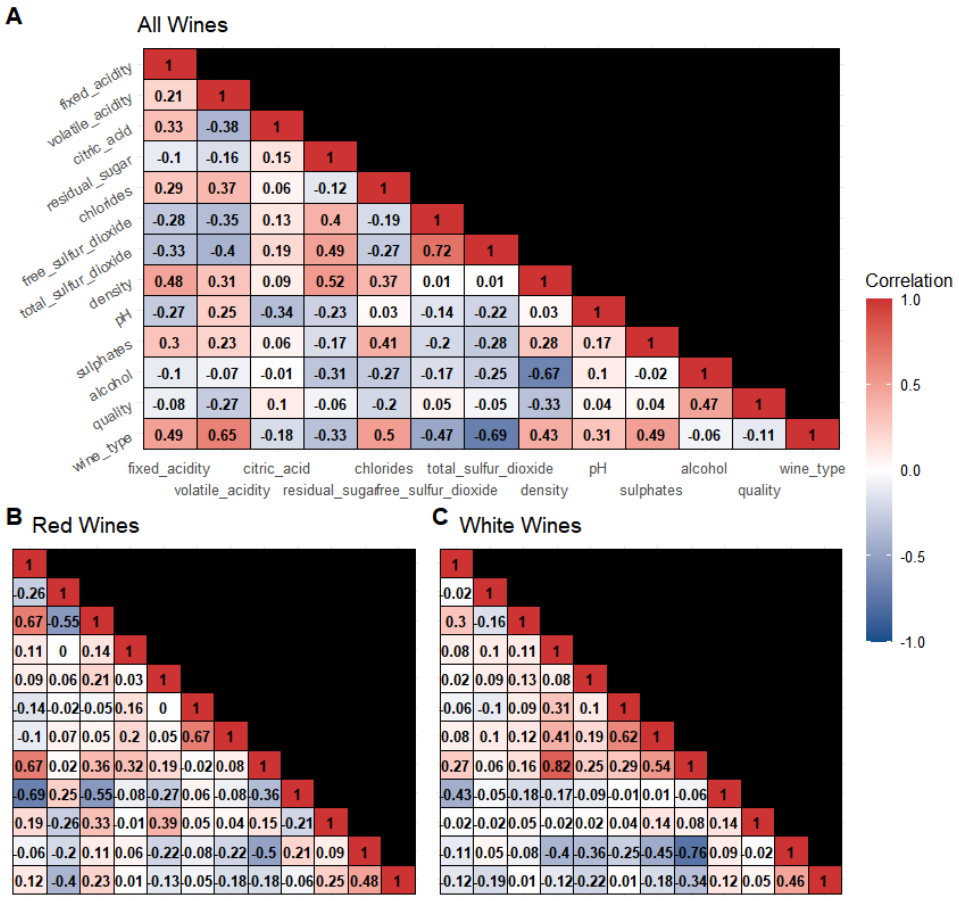


Figure 2: Correlation heatmaps of predictors. (A) All wines combined show few useful correlations due. (B) Red wines only, highlighting acidity and density relationships. (C) White wines only, highlighting the negative relationship between residual sugar and density. Note: Sub-figures (B) and (C) share the same axes labels as (A), except for wine_type, which is excluded.

Heatmaps stratified by quality groups offered little additional insights (and were thus not included), facing similar issues as the all-wine heatmap. The only consistent quality-related signal is the positive correlation between alcohol content and quality (r $\approx$ 0.46–0.48), evident across all subfigures in Figure 2, which directly informs our second research question on identifying indicators of wine quality.

4 Principal Component Analysis

Because variables have different units and magnitudes, we conducted PCA on centered and scaled variables. When selecting important components, using the eigenvalue ratio method, we found no stable components (ratio > 2) dominating the explained variance, so we opted for Kaiser's criterion (eigenvalues > 1), which suggested retaining the first three principal components explaining 64.1% of total variance (Figure 3). Resulting eigenvalues

from PCA show that the first two components explain 49.7% of the total variance, so visualizations focus on PC1 and PC2.

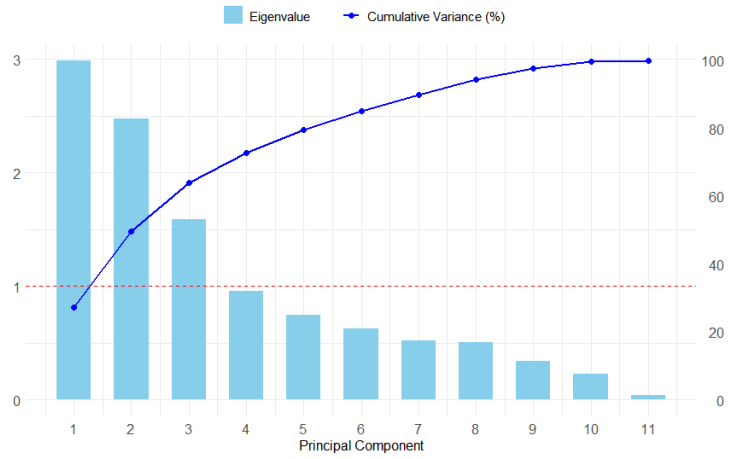


Figure 3: Scree plot of principal components. Bars show eigenvalues corresponding to each component, representing the amount of variance explained in each direction. The cumulative variance explained is shown by the dark blue curve. The dotted red line marks Kaiser's criterion (eigenvalues > 1), which guided our decision to retain the first three components, collectively explaining 64.1% of the total variance.

4.1 Interpretations of PC Loadings

PC1 largely confirms the patterns observed in the correlation heatmaps (Figure 2): red wines have positive loadings for fixed acidity, volatile acidity, chlorides, sulphates, and partially pH. Meanwhile, white wines are characterized by higher loadings for sulfur dioxides, citric acid, and residual sugar.

On the other hand, PC2 captures secondary contributions, primarily from alcohol and the previously noted negative correlation between alcohol and density.

4.2 PCA Score Plots and Group Differences

To further explore quality distinctions, we plotted PCA scores grouped jointly by wine type and quality group in Figure 4B, including only "Bad" and "Great" wines for visual clarity and omitting the "Average" group. The centroids of bad and great wines differ meaningfully in PC space, with great wines generally positioned higher on both PC1 and PC2 axes, consistent with our earlier hypothesis that PC2 captures quality-related variance.

Moreover, higher-quality wines exhibit lower variance in PC space, reinforcing the observation that they are more balanced. Note that there is an exception that average white wines have lower variance along PC2 than great white wines.

Finally, an interesting remark about the Minho wine region: producers prioritize white wines, producing approximately three times more white than red wines. This aligns with the proportion of high-quality wines, as great white wines are nearly four times as common as great red wines in this dataset (4.37% vs. 1.25%).

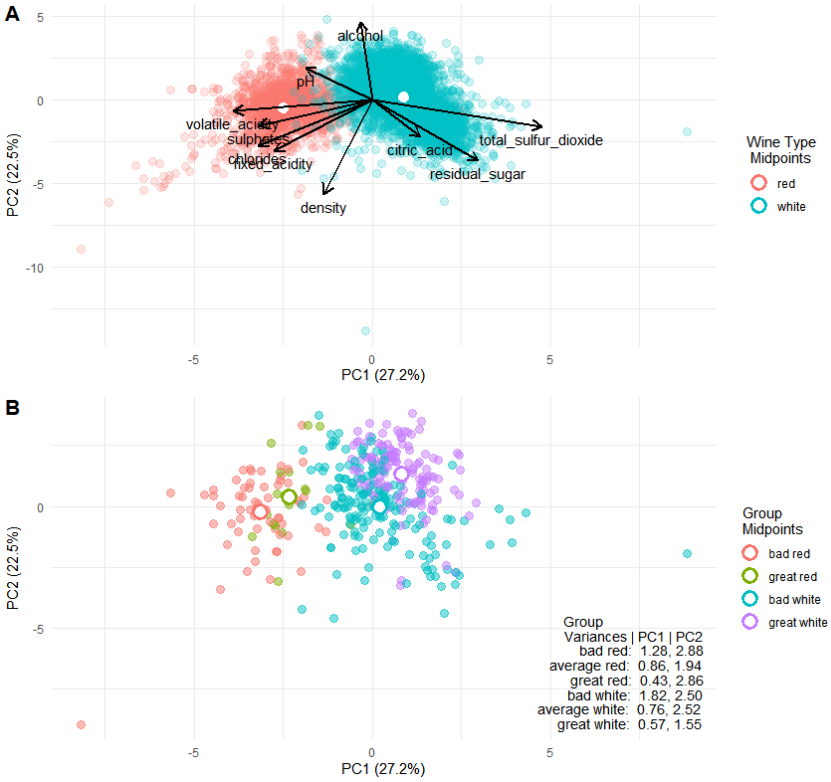


Figure 4: PCA score biplots of PC1 and PC2. (A) Shows the clear separation between red and white wines along PC1. (B) Further stratifying by wine quality and type highlights inter-group distinctions: higher-quality wines tend to occupy higher values on both principal component axes and exhibit lower variance (values indicated in the bottom-right corner of panel B), indicating that they are more balanced.

5 Discussion

Overall, both the type of wine and the quality of the wine have been found to have an underlying structure for our research questions.

- Wine type is strongly captured by PC1, reflecting chemical compositions consistently observed in oenology. This differentiation in white and red wines is evident in all the heatmaps (Figure 2) and the PCA biplot (Figure 4A).
- Wine quality is primarily associated with alcohol content, with additional subtler factors also contributing, as shown in Figure 2 and Figure 4B.
- Inter-group variances in Figure 4B indicate that higher-quality wines tend to be more chemically consistent, whereas lower-quality wines exhibit larger imbalances.

Together, EDA, correlation analysis, and PCA show that physicochemical measurements contain meaningful, though incomplete, information about wine qualtiy.

6 Limitations

Although this dataset provides a broad overview of a large collection of wines and yields valuable insights, it lacks some variables that may account for the unpredictability and

variance in quality ratings. Due to privacy and logistical constraints, the original data collectors were unable to include information on wine brand, selling price, or most importantly, grape varietals Cortez et al. (2009). This is a notable limitation because chemical profiles can vary substantially across varietals; for example, Syrah grapes are known to absorb more chlorides than others Coli et al. (2015). In addition, while PCA is generally robust to non-normality, the heavy-tailed distributions present in several variables may still influence the results, even after scaling, log scaling, or other transformations of the data should be considered in the future.

7 Conclusion

This analysis demonstrates that physicochemical measurements can effectively distinguish red from white wines. However, these features alone cannot fully explain or predict wine quality, reflecting the complexity of the human palate. Future studies incorporating additional contextual information, particularly grape varietals, would enable more comprehensive modeling of wine quality and help control for confounding effects that are unobservable in the current dataset.

8 AI Declaration

AI assisted with grammar in the report and with some code, mostly making plots.

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